

ITERATIVE METHODS FOR THE NON-LTE TRANSFER OF POLARIZED RADIATION: RESONANCE LINE POLARIZATION IN ONE-DIMENSIONAL ATMOSPHERES

JAVIER TRUJILLO BUENO AND RAFAEL MANSO SAINZ

Instituto de Astrofísica de Canarias, 38200 La Laguna, Tenerife, Spain; jtb@ll.iac.es

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ABSTRACT

This paper shows how to generalize to non-LTE polarization transfer some operator splitting methods that were originally developed for solving unpolarized transfer problems. These are the Jacobi-based accelerated Λ -iteration (ALI) method of Olson, Auer, & Buchler and the iterative schemes based on Gauss-Seidel and successive overrelaxation (SOR) iteration of Trujillo Bueno and Fabiani Bendicho. The theoretical framework chosen for the formulation of polarization transfer problems is the quantum electrodynamics (QED) theory of Landi Degl’Innocenti, which specifies the excitation state of the atoms in terms of the irreducible tensor components of the atomic density matrix. This first paper establishes the grounds of our numerical approach to non-LTE polarization transfer by concentrating on the standard case of scattering line polarization in a gas of two-level atoms, including the Hanle effect due to a weak microturbulent and isotropic magnetic field. We begin demonstrating that the well-known Λ -iteration method leads to the self-consistent solution of this type of problem if one initializes using the “exact” solution corresponding to the unpolarized case. We show then how the above-mentioned splitting methods can be easily derived from this simple Λ -iteration scheme. We show that our SOR method is 10 times faster than the Jacobi-based ALI method, while our implementation of the Gauss-Seidel method is 4 times faster. These iterative schemes lead to the self-consistent solution independently of the chosen initialization. The convergence rate of these iterative methods is very high; they do not require either the construction or the inversion of any matrix, and the computing time per iteration is similar to that of the Λ -iteration method.

Subject headings: line: profiles — methods: numerical — polarization — radiative transfer — scattering — stars: atmospheres — Sun: atmosphere

1. INTRODUCTION

Observations of linear polarization close to the solar limb performed with novel spectropolarimeters (Povel 1995) that allow the detection of very low amplitude polarization signals are leading to the discovery of unexpected new phenomena and physical processes (Stenflo & Keller 1996, 1997; Trujillo Bueno & Landi Degl’Innocenti 1997; Landi Degl’Innocenti 1998; Trujillo Bueno 1999). In order to exploit the rich diagnostic potential of these and future polarimetric observations, we need to be able to perform detailed numerical simulations of the generation and transfer of polarized radiation in magnetized atmospheres taking properly into account scattering processes in multilevel atomic models. This is also of interest in other astrophysical contexts (e.g., symbiotic stars, winds from hot stars, and active galactic nuclei) where scattering polarization may have a diagnostic value (see, e.g., Schmid 1995; Ignace, Nordieck, & Cassinelli 1997; Hillier 1996). To this end, it is first imperative to develop fast iterative methods suitable for the solution of general non-LTE polarization transfer problems in one-, two-, and three-dimensional geometries.

This paper is the first of a series that has this ambitious goal as its aim. It deals with the simplest two-level atom non-LTE polarization transfer problem, i.e., with scattering line polarization in a one-dimensional, plane-parallel stellar atmosphere that is permeated by a weak microturbulent and isotropic magnetic field. Our aim is to develop highly efficient iterative methods for the self-consistent solution of the relevant kinetic and radiative transfer equations. As we shall show, our iterative methods for the transfer of polarized radiation are capable of yielding rapidly, at each spatial point, the quantities that fully specify the atomic

excitation, i.e., the irreducible tensor components of the atomic density matrix (see Landi Degl’Innocenti 1996; Bommier 1996). For each atomic level, the elements of its corresponding density matrix contain information about the populations of its atomic sublevels and about the degree of quantum interference (or coherence) among the sublevels themselves. The sought values of these density matrix elements have to be *consistent* with the polarization properties of the radiation field produced by the assumed chemical species in the given medium. Once this self-consistent atomic excitation is known along the line of sight, it is straightforward to solve the transfer equation for the Stokes parameters in order to calculate the emergent polarization profiles that are to be compared with spectropolarimetric observations.

In this particular resonance line polarization problem that we are considering here, the only nonvanishing Stokes parameters are the intensity (I) and the linear polarization component (Q). The circular polarization (V) and the other linear polarization component (U) are zero because of the problem’s assumptions and the rotational symmetry of the radiation field with respect to the normal to the stellar surface. Also, only two density matrix elements suffice to determine here the atomic excitation, namely, the overall population of the upper level and a coefficient (called the upper-level atomic alignment) that measures the degree of imbalance in the populations of the upper-level sublevels (see Landi Degl’Innocenti, Bommier, & Sahal-Bréchet 1990).

This problem is also characterized by the neglect of stimulated emission and of any lower-level atomic alignment (Landi Degl’Innocenti 1996). These two assumptions

imply that the corresponding equations are *linear*, as is the case with the standard *unpolarized* two-level atom problem (see Mihalas 1978). In fact, this particular two-level atom resonance line polarization case offers the easiest conceptual transition between unpolarized and polarized non-LTE transfer. Therefore, it constitutes the starting point in our systematic description of the iterative methods for the non-LTE transfer of polarized radiation that we have been developing over the last few years.

Before entering into details, we should mention our opinion about the state of the art in this field of research. Concerning the development of theories for the generation and transfer of polarized radiation, we believe that the most rigorous (and suitable) theoretical framework to work with is the one developed by Landi Degl'Innocenti (1983), which is based on the irreducible tensor components of the atomic density matrix (see also Bommier & Sahal-Br         1978). Although this formalism is certainly more cumbersome than the one used in other more phenomenological theories (see, e.g., Stenflo 1994), it has the advantage of allowing one to formulate with confidence polarization transfer problems of increasing complication, taking properly into account a variety of relevant (and subtle) physical mechanisms that play an important role in producing the observed polarization (see Trujillo Bueno & Landi Degl'Innocenti 1997; Landi Degl'Innocenti 1998; Trujillo Bueno 1999). The main criticism of this quantum electrodynamics (QED) theory is that it is based on the approximation of complete frequency redistribution, i.e., on an assumption that is not adequate for modeling the polarization of several diagnostically important spectral lines. Fortunately, some very recent works have successfully started to incorporate the effects of partial redistribution into the framework of the density matrix formalism (see Landi Degl'Innocenti, Landi Degl'Innocenti, & Landolfi 1997; Bommier 1997a, 1997b).

For the line polarization problem considered here, the equations obtained with the density matrix theory (see Landi Degl'Innocenti et al. 1990) are equivalent to those found via the application of some phenomenological theories (see, e.g., Rees 1978; Stenflo 1994; Ivanov 1995). However, all our work on the subject of non-LTE polarization transfer will be presented within the framework of the density matrix theory because this has allowed us to treat more complex polarization transfer problems that cannot be conveniently treated otherwise (see Trujillo Bueno & Landi Degl'Innocenti 1997; Trujillo Bueno 1999).

As in the standard unpolarized transfer case (see Mihalas 1978), the non-LTE polarization transfer problem requires solving *self-consistently* two coupled sets of equations: the statistical equilibrium (SE) equations (which describe the excitation and ionization states) and the radiative transfer (RT) equations (which govern the behavior of the radiation field). The combination leads to a coupled set of equations of the form

$$\mathbf{A}x = \mathbf{b},$$

where $\mathbf{A}$ is an operator which depends on the collisional and radiative rates, $\mathbf{b}$ is a known vector, and x is the *unknown*, i.e., the vector formed by the density matrix elements at all the spatial grid points. A transfer problem is *linear* if the operator $\mathbf{A}$ does not depend on the unknown x . As indicated previously, for this one needs to choose a two-level atomic model neglecting stimulated emission processes and to assume that there is no atomic polarization in the

lower level, i.e., that there are no population imbalances among the lower-level sublevels and no coherences among them. Apart from the resonance line polarization problem considered in this paper, other examples of two-level atom problems that are linear are the Hanle effect (see, e.g., Faurobert-Scholl 1996) and the Zeeman line transfer case (see, e.g., Bommier & Landi Degl'Innocenti 1996; Trujillo Bueno & Landi Degl'Innocenti 1996a). All these linear problems can in principle be solved by applying *direct* matrix inversion methods (see, e.g., Faurobert-Scholl 1991; Bommier, Landi Degl'Innocenti, & Sahal-Br         1991; Bommier & Landi Degl'Innocenti 1996). However, these methods are of little practical interest because the computing time needed to build the operator $\mathbf{A}$ is very large, apart from the fact that the computational work required to invert it scales as the cube of the number of unknowns. This implies that, for large-scale problems (e.g., in two- and three-dimensional geometries), the solution itself soon becomes polluted with rounding errors.

The alternative is iterative methods. An example is the Jacobi-based accelerated Λ -iteration (ALI) method (see Olson, Auer, & Buchler 1986) that has been successfully used for more than a decade for solving unpolarized transfer problems, for both two-level and multilevel atomic models. This method has been applied by Trujillo Bueno & Landi Degl'Innocenti (1996) to the non-LTE Zeeman line transfer case where all of the four Stokes parameters are coupled together because of the appearance in the transfer equation of the full 4×4 absorption matrix. More recently, Faurobert-Scholl, Frisch, & Nagendra (1997) and Nagendra, Frisch, & Faurobert (1998) applied the Jacobi-based ALI technique to solve the resonance line polarization and the Hanle effect problem in one-dimensional media, as formulated using Ivanov's (1995) formalism. Finally, the Jacobi-based ALI method has also been successfully generalized by Manso Sainz & Trujillo Bueno (1999) to the resonance line polarization and the Hanle effect in one-, two-, and three-dimensional media, as formulated using the density matrix theory.

Iterative schemes of a superior type are the Gauss-Seidel (GS) based methods of Trujillo Bueno & Fabiani Bendicho (1995). One of the strong points of this paper is to describe in some detail how these GS-based methods can be generalized to non-LTE polarization transfer because they give an improvement of an order of magnitude in the computational work with respect to the Jacobi-based ALI method. In our presentation below we will start showing how the well-known Λ -iteration method can be applied to polarization transfer, in order to show then how the Jacobi, Gauss-Seidel, and successive overrelaxation (SOR) methods can be easily derived when certain modifications are introduced into the Λ -iteration equations.

In general, however, the operator $\mathbf{A}$ depends on the unknown x , i.e., the non-LTE polarization transfer problem is *nonlinear*. This is because the absorption matrix and the emission vector that appear in the Stokes vector transfer equation depend on the atomic density matrix elements. Thus, the Stokes parameters (and the radiative rates on which the elements of the operator $\mathbf{A}$ depend) also depend on them. Thus, the quantities to be calculated (i.e., the density matrix elements) depend nonlinearly on themselves via the transfer equation. The solution of nonlinear problems necessarily requires the application of iterative methods. Here, at each iterative step, one has to manage to

set up and solve a suitable linear system of equations whose solution leads to approximate corrections to the unknowns (see Socas Navarro & Trujillo Bueno 1997). For the above-mentioned reasons, iterative schemes that require the construction and inversion of large matrices are useless. Thus, the aim is to develop effective iterative methods capable of solving nonlinear polarization transfer problems without having to build and invert large matrices at each iterative step.

An example of a nonlinear polarization transfer problem that can be rigorously formulated using the density matrix theory is the case of resonance line polarization in a gas of two-level atoms, but taking into account the possibility of atomic polarization in the lower level. A Jacobi-based method for the solution of this *nonlinear* problem has been applied by Trujillo Bueno & Landi Degl'Innocenti (1996b, 1997) in their investigation on linear polarization due to lower-level depopulation pumping processes. An attractive feature of this method (see also Trujillo Bueno 1999) is that it reduces to the Jacobi-based one presented in this paper when the lower-level atomic polarization is neglected.

The situation is complicated dramatically when one assumes a multilevel atomic model instead of a two-level one. To give a feeling of the added complexity, it suffices to take into account that to each level of total angular momentum J correspond $(2J + 1)^2$ density matrix elements. This applies to cases with arbitrary magnetic fields, but neglecting hyperfine structure effects. The inclusion of hyperfine structure complicates the equations still further. In summary, the large number of unknowns and the fact that the radiation field now has to be described in terms of the Stokes vector make the non-LTE polarization transfer problem both a particularly complex and a fascinating research field.

Certainly the issue of the numerical solution of this type of problem is formidable. The aim is to develop accurate formal solution methods of the Stokes vector transfer equation and powerful iterative schemes capable of yielding rapidly at each spatial grid point the self-consistent values of the atomic density matrix elements. We are now in a stage in which we have already developed suitable formal solvers and three types of iterative methods for radiative transfer applications that are capable of solving a variety of polarization transfer problems of increasing complication (see Trujillo Bueno 1998, 1999). Of particular interest are the Hanle effect in one-, two-, and three-dimensional geometries, the Zeeman line transfer case, and the resonance line polarization problem taking into account ground-level atomic polarization. These problems are indeed substantially more complicated than the one considered in the present paper. However, since our numerical approach has been presented only in several workshop talks (Trujillo Bueno & Landi Degl'Innocenti 1996a, 1996b, 1997; Trujillo Bueno 1998, 1999; Manso Sainz & Trujillo Bueno 1999; Fabiani Bendicho & Trujillo Bueno 1999), and no detailed description of the generalization of our iterative methods to polarization transfer has been published up to now, we restrict ourselves in this first journal paper to consider the simplest two-level atom non-LTE polarization transfer case. The aim of this first paper is thus to establish the grounds and to initiate a systematic description of our numerical approach to non-LTE polarized transfer. Forthcoming papers will then consider other more complex linear two-level atom problems and the generalization of our iter-

ative schemes to the nonlinear case, for both two-level and multilevel atomic systems.

The outline of this paper is as follows. Section 2 deals with the formulation of the problem, presenting a brief review of the SE and RT equations that can be obtained applying the density matrix theory of Landi Degl'Innocenti (1983, 1984). The Appendix shows how to obtain the SE equations including the effect of a microturbulent and isotropic weak magnetic field. Section 3 considers how three iterative methods (Jacobi, Gauss-Seidel, and SOR) can be applied to this resonance line polarization case. Here the emphasis is on demonstrating the suitability of the Gauss-Seidel-based methods of Trujillo Bueno & Fabiani Bendicho (1995). Section 4 is dedicated to illustrating with some examples the reliability of the methods, and § 5 gives our conclusions.

2. FORMULATION OF THE PROBLEM

We consider the standard two-level atom resonance line polarization problem in a one-dimensional, plane-parallel, static stellar atmosphere in the presence of a weak microturbulent and isotropic magnetic field that influences the emergent polarization via the Hanle effect. We use the symbols J_l and J_u to denote the total angular momentum of the lower and upper levels, respectively. In order to keep the problem linear, stimulated emission processes and atomic polarization in the lower level are neglected (Landi Degl'Innocenti 1996). The assumption of neglecting atomic polarization in the lower level implies that the populations of the lower-level sublevels are all equal. We point out that atomic polarization can be divided into atomic *orientation* and atomic *alignment*.

We would have atomic orientation if the populations of the sublevels (pertaining to the same level) with magnetic quantum numbers M and $-M$ were different. Atomic orientation is indeed zero in this standard resonance line polarization case, both for the lower and upper levels, because it is assumed that the radiation illuminating the boundaries has no circular polarization (see the Appendix; see also the paper by Trujillo Bueno et al. 1993 for a situation where atomic orientation may come into play).

However, atomic alignment is, in general, nonzero because the populations of the sublevels (pertaining to the same level) with $|M| \neq |M'|$ can indeed be different. This can happen whenever the radiation field that induces radiative transitions in the atom is anisotropic and/or polarized. For instance, if an unpolarized light beam propagating along the direction chosen as the quantization axis illuminates a gas of two-level atoms with $J_l = 0$ and $J_u = 1$, only the transitions corresponding to $\Delta M = \pm 1$ are effective, so that no transitions occur to the $M = 0$ sublevel of the upper level. Thus, in the absence of any relaxation mechanisms, the upper-level sublevels with $M = 1$ and $M = -1$ would be more populated than the $M = 0$ sublevel.

The assumption of neglecting atomic alignment in the lower level is only strictly true for line transitions having $J_l = 0$ or $J_l = \frac{1}{2}$ and for atoms without hyperfine structure. For this approximation to be adequate for line transitions with other J_l values, one has to assume that the effect of elastic depolarizing collisions is sufficiently strong to equalize the populations of the magnetic sublevels pertaining to the same level. For a given collisional depolarizing rate the efficiency of depolarizing collisions is larger, the larger the lifetime of the atomic level. Since for a two-level model atom

the lower-level lifetime is larger than the upper-level lifetime, elastic collisions are generally more efficient in reducing the lower-level alignment than the upper-level one. Thus, the assumption of neglecting ground-level polarization can be considered, a priori, as a reasonable approximation that is introduced mainly to keep the problem linear. For recent investigations that take properly into account the physical ingredient of ground-level atomic polarization we refer the reader to the papers by Trujillo Bueno & Landi Degl'Innocenti (1997), Landi Degl'Innocenti (1998), and Trujillo Bueno (1999).

In summary, in the standard resonance line polarization two-level atom problem that we are considering here, population imbalances are only assumed to occur among the sublevels of the upper level. The overall population of the lower level (which determines the opacity stratification) is assumed to be fixed. To specify the excitation state at each spatial grid point, it is enough to consider two density matrix elements: ρ_0^0 and ρ_0^2 . The first one measures the overall population of the upper level, while ρ_0^2 is the so-called alignment coefficient which quantifies the degree of imbalance in the populations of the upper-level sublevels. For instance, for a level with $J = 1$, $\rho_0^0 = (n_1 + n_0 + n_{-1})/\sqrt{3}$ and $\rho_0^2 = (n_1 - 2n_0 + n_{-1})/\sqrt{6}$, where n_i (with $i = 1, 0$, and -1) are the populations of the three magnetic sublevels. These two ρ_0^K elements are the irreducible tensor components of the density matrix that correspond to this one-dimensional resonance line polarization problem, and they are the two unknowns whose self-consistent values at each spatial grid point are sought. It is interesting to point out that in the above example where two-level atoms with $J_l = 0$ and $J_u = 1$ were assumed to be illuminated by a unidirectional unpolarized light beam, we have $\rho_0^2 > 0$. Clearly, the amount of this atomic alignment in a stellar atmospheric environment is significantly smaller than in such an academic case of a unidirectional light beam, owing to the relaxation mechanisms present in a stellar atmosphere (e.g., depolarizing collisions and magnetic fields) and to the much weaker degree of its radiation field's anisotropy.

The application of the density matrix theory of Landi Degl'Innocenti (1983) leads to the following set of SE equations (see the Appendix):

$$S_0^0 = \frac{2h\nu^3}{c^2} \frac{2J_l + 1}{\sqrt{2J_u + 1}} \rho_0^0 = (1 - \epsilon)\bar{J}_0^0 + \epsilon B_\nu, \quad (1)$$

$$S_0^2 = \frac{2h\nu^3}{c^2} \frac{2J_l + 1}{\sqrt{2J_u + 1}} \rho_0^2 = \mathcal{H}^{(2)} \frac{1 - \epsilon}{1 + \delta^{(2)}(1 - \epsilon)} w_{J_u J_l}^{(2)} \bar{J}_0^2, \quad (2)$$

where ν is the line frequency, $\mathcal{H}^{(2)}$ is a Hanle depolarization factor that varies between 1 (for $\nu_L/A_{ul} = 0$; ν_L is the Larmor frequency, which is proportional to the intensity of the magnetic field) and 0.2 (for $\nu_L/A_{ul} \rightarrow \infty$), ϵ is the collisional destruction probability due to *inelastic* collisions, $\delta^{(2)}$ is the collisional depolarizing rate due to *elastic* collisions measured in units of the Einstein A_{ul} coefficient, $w_{J_u J_l}^{(2)}$ is a coefficient whose value depends on the total angular momentum of the lower and upper levels (e.g., $w_{J_u J_l}^{(2)} = 1$ for a $J_l = 0 \rightarrow J_u = 1$ transition), $\bar{J}_0^0$ and $\bar{J}_0^2$ are two radiation field tensors that we will consider below, and ρ_0^0 and ρ_0^2 are the density matrix elements of the upper level normalized to the overall population of the ground level. In these equa-

tions we have also defined the quantities S_0^0 and S_0^2 , which are the source function tensors introduced by Landi Degl'Innocenti & Bommier (1994). We point out here that in all the illustrative calculations shown below we have taken $2h\nu^3/c^2 = 1$, $\mathcal{H}^{(2)} = 1$, $J_l = 0$ and $J_u = 1$.

Since the atmosphere is assumed to be one-dimensional and the microturbulent weak magnetic field has no preferred direction, the radiation field has rotational symmetry with respect to the vertical to the stellar atmosphere and the only nonvanishing Stokes parameters are I and Q . The transfer equations for these Stokes parameters are

$$\frac{d}{d\tau_x} I = I - S_I, \quad (3)$$

$$\frac{d}{d\tau_x} Q = Q - S_Q, \quad (4)$$

where τ_x is the monochromatic optical depth along the ray (see below), with x the frequency distance from line center measured in units of the Doppler width, and S_I and S_Q are the source function components of the I and Q Stokes parameters, respectively. Assuming an unpolarized background continuum, their expressions are

$$S_I = \frac{\chi_l \phi_x S_I^{\text{line}} + \chi_c S_c}{\chi_l \phi_x + \chi_c} = r_x S_I^{\text{line}} + (1 - r_x) S_c, \quad (5)$$

$$S_Q = \frac{\chi_l \phi_x S_Q^{\text{line}}}{\chi_l \phi_x + \chi_c} = r_x S_Q^{\text{line}}, \quad (6)$$

where S_c is the continuum source function that we will assume equal to the Planck function, ϕ_x is the absorption profile, χ_l and χ_c are the line-integrated and continuum opacities, respectively, $d\tau_x = -(\chi_c + \chi_l \phi_x) ds$ (with s the geometrical distance along the ray), and

$$r_x = \frac{\chi_l \phi_x}{\chi_l \phi_x + \chi_c}. \quad (7)$$

The line contributions S_I^{line} and S_Q^{line} depend on the values of ρ_0^0 and ρ_0^2 (see Landi Degl'Innocenti 1984):

$$S_I^{\text{line}} = \frac{2h\nu^3}{c^2} \frac{2J_l + 1}{\sqrt{2J_u + 1}} \left[\rho_0^0 + \frac{w_{J_u J_l}^{(2)}}{2\sqrt{2}} (3\mu^2 - 1) \rho_0^2 \right], \quad (8)$$

$$S_Q^{\text{line}} = \frac{2h\nu^3}{c^2} \frac{2J_l + 1}{\sqrt{2J_u + 1}} \frac{3w_{J_u J_l}^{(2)}}{2\sqrt{2}} (\mu^2 - 1) \rho_0^2. \quad (9)$$

As mentioned above $\bar{J}_0^0$ and $\bar{J}_0^2$ are two radiation field tensors that enter the statistical equilibrium equations. They are given by the following weighted integrals over frequency and direction of the Stokes I and Q parameters (see Landi Degl'Innocenti 1984):

$$\bar{J}_0^0 = \frac{1}{2} \int dx \phi(x) \int_{-1}^1 d\mu I(x, \mu), \quad (10)$$

and

$$\begin{aligned} \bar{J}_0^2 = & \frac{1}{4\sqrt{2}} \int dx \phi(x) \int_{-1}^1 d\mu [(3\mu^2 - 1)I(x, \mu) \\ & + 3(\mu^2 - 1)Q(x, \mu)], \end{aligned} \quad (11)$$

where $\mu = \cos \theta$ (with θ the angle between the ray path and the vertical).

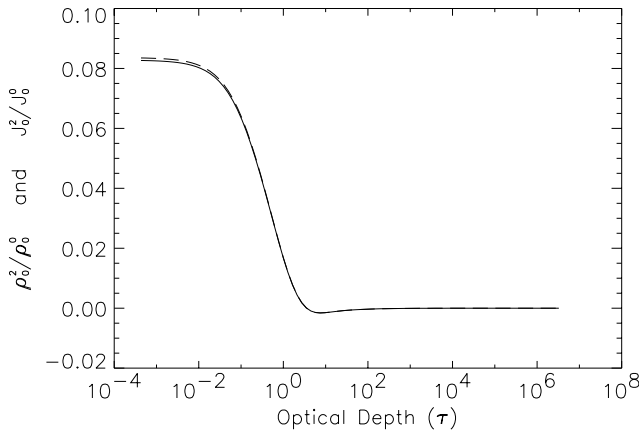


FIG. 1.—Variation with the line-integrated optical depth of ρ_0^2/ρ_0^2 (solid line) and of $\bar{J}_0^2/\bar{J}_0^2$ (dashed line), for a two-level atom calculation in isothermal atmospheres without any background continuum and for $\epsilon = 10^{-4}$.

The dashed line of Figure 1, obtained through the self-consistent solution of the set of equations (1) and (2) assuming $\epsilon = 10^{-4}$ and $\delta^{(2)} = 0$, shows the variation of ρ_0^2/ρ_0^2 and of $\bar{J}_0^2/\bar{J}_0^2$ versus line-integrated optical depth. Note that the maximum absolute value of $\bar{J}_0^2/\bar{J}_0^2$ occurs at the surface, where it is about 8%, and that in the absence of depolarizing collisions $\rho_0^2/\rho_0^2 \sim \bar{J}_0^2/\bar{J}_0^2$, as expected from equations (1) and (2).

As indicated by equation (10), the calculation of $\bar{J}_0^2$ only requires knowledge of the Stokes I -parameter. Note that $\bar{J}_0^2$ would be nonzero even if this I -parameter were isotropic. However, $\bar{J}_0^2$ (cf. eq. [11]) depends on both, the Stokes I and Q parameters and is only nonnegligible in the surface atmospheric layers where the anisotropy of the radiation field is significant. Since the fractional linear polarization Q/I is at most of the order of a few percent, the radiation field tensor $\bar{J}_0^2$ quantifies mainly the anisotropy of the specific intensity of the radiation field. It is this anisotropic illumination of the atoms that leads to significant $\bar{J}_0^2$ values (typically $\bar{J}_0^2/\bar{J}_0^2$ values of the order of 8% close to the surface) and, as shown by equation (2), this produces population imbalances among the sublevels of the atomic levels. This atomic alignment leads to the generation of linear polarization signals at each point in the stellar atmosphere, signals that are later modified via transfer processes in the stellar plasma.

3. ITERATIVE METHODS OF SOLUTION

The simplest iterative scheme one might think of to find at each spatial grid point the self-consistent values of ρ_0^0 and ρ_0^2 is Λ -iteration. Given an estimate $\rho_0^{0\text{old}}$ and $\rho_0^{2\text{old}}$ of the density matrix elements (or, alternatively, of S_0^0 and S_0^2), one can calculate S_I and S_Q from equations (5) and (6). One can then perform a formal solution of the transfer equations (3) and (4) to obtain I and Q at each spatial grid point for all the frequency-angle points of the quadrature chosen for solving numerically the integrals that define $\bar{J}_0^0$ and $\bar{J}_0^2$. We agree to denote the radiation field tensors so calculated with the symbols $\bar{J}_0^{0\text{old}}$ and $\bar{J}_0^{2\text{old}}$, in order to indicate that they were calculated from the current estimates $\rho_0^{0\text{old}}$ and $\rho_0^{2\text{old}}$ of the density matrix elements. Introducing these “old” radiation field tensor values into the SE equations (1) and (2), one can obtain “new” estimates $\rho_0^{0\text{new}}$ and $\rho_0^{2\text{new}}$. Iterating this process is nothing but the well-known Λ -iteration method. We can also express this iterative scheme with

equations as follows:

$$S_0^{0\text{new}} = (1 - \epsilon)\bar{J}_0^{0\text{old}} + \epsilon B_\nu, \quad (12)$$

$$S_0^{2\text{new}} = \mathcal{H}^{(2)} \frac{1 - \epsilon}{1 + \delta^{(2)}(1 - \epsilon)} w_{J_u J_l}^{(2)} \bar{J}_0^{2\text{old}}. \quad (13)$$

Figure 2 shows an example of the convergence properties of this Λ -iteration scheme. It gives the variation with the iteration number of the relative *true* error corresponding to the ρ_0^0 and ρ_0^2 values at the surface grid point. The only difference is that the calculation corresponding to the upper panels was initialized using the LTE ρ -values, while that corresponding to the lower panels was performed using as initial estimate the *self-consistent* density matrix element values corresponding to the particular case in which polarization is neglected, i.e., using $\rho_0^2 = 0$ and the self-consistent ρ_0^0 values that are obtained by solving equation (1) alone assuming that $\rho_0^2 = 0$ in equation (8). The conclusion is that the Λ -iteration method is indeed able to lead to the “exact” solution if one initializes as in the calculation of the lower panels of Figure 2, i.e., using the self-consistent solution corresponding to the standard two-level atom problem in which polarization phenomena are disregarded. (Note from the values at the iteration number zero in the two lower panels of Fig. 2 that the *true* error associated with the ρ_0^0 value of this initialization is of the order of 1%, i.e., that this initialization is very good for ρ_0^0 .) This is, after all, what one would expect from an inspection of the problem’s equations and from the fact that $\rho_0^2/\rho_0^0 \sim \bar{J}_0^2/\bar{J}_0^0 \ll 1$.

Although the idea of solving non-LTE polarization transfer problems applying the Λ -iteration method, but initializing as we have previously indicated, may turn out to be very useful for solving some particularly complex problems (e.g., the modeling of scattering polarization signals in spectral lines using multilevel atomic models), it is crucial to develop more powerful methods capable of yielding rapidly the self-consistent ρ -values independently of the chosen initialization. In order to develop iterative schemes that converge much faster than the Λ -iteration method, but still sharing its simplicity, we are going to account *implicitly* for some information about the “new” values of the ρ -elements that enter the expressions of the radiation field tensors $\bar{J}_0^0$ and $\bar{J}_0^2$ via the dependence of S_I and S_Q on ρ_0^0 and ρ_0^2 .

3.1. ALI or Jacobi Iteration

The calculation of the radiation field tensors $\bar{J}_0^0$ and $\bar{J}_0^2$ requires the knowledge of the Stokes parameters I and Q for each frequency-angle point of the chosen numerical quadrature. For a given discrete spatial grid of NP points the formal solution of the transfer equations (3) and (4) can be symbolically represented as

$$I_{x\Omega} = \Lambda_{x\Omega}[S_I] + T_{x\Omega}^I, \quad (14)$$

$$Q_{x\Omega} = \Lambda_{x\Omega}[S_Q] + T_{x\Omega}^Q, \quad (15)$$

where $T_{x\Omega}^X$ (with X either I or Q) gives the transmitted Stokes X -parameter due to the incident radiation at the boundary, and $\Lambda_{x\Omega}$ is an $NP \times NP$ operator whose elements depend on the optical distances between the grid points. A suitable formal solution method for the numerical solution of the two independent scalar transfer equations (3) and (4) is the short-characteristics (SC) method (Kunasz & Auer 1988; Auer & Paletou 1994). This method is based on the assumption that the source function varies parabolically

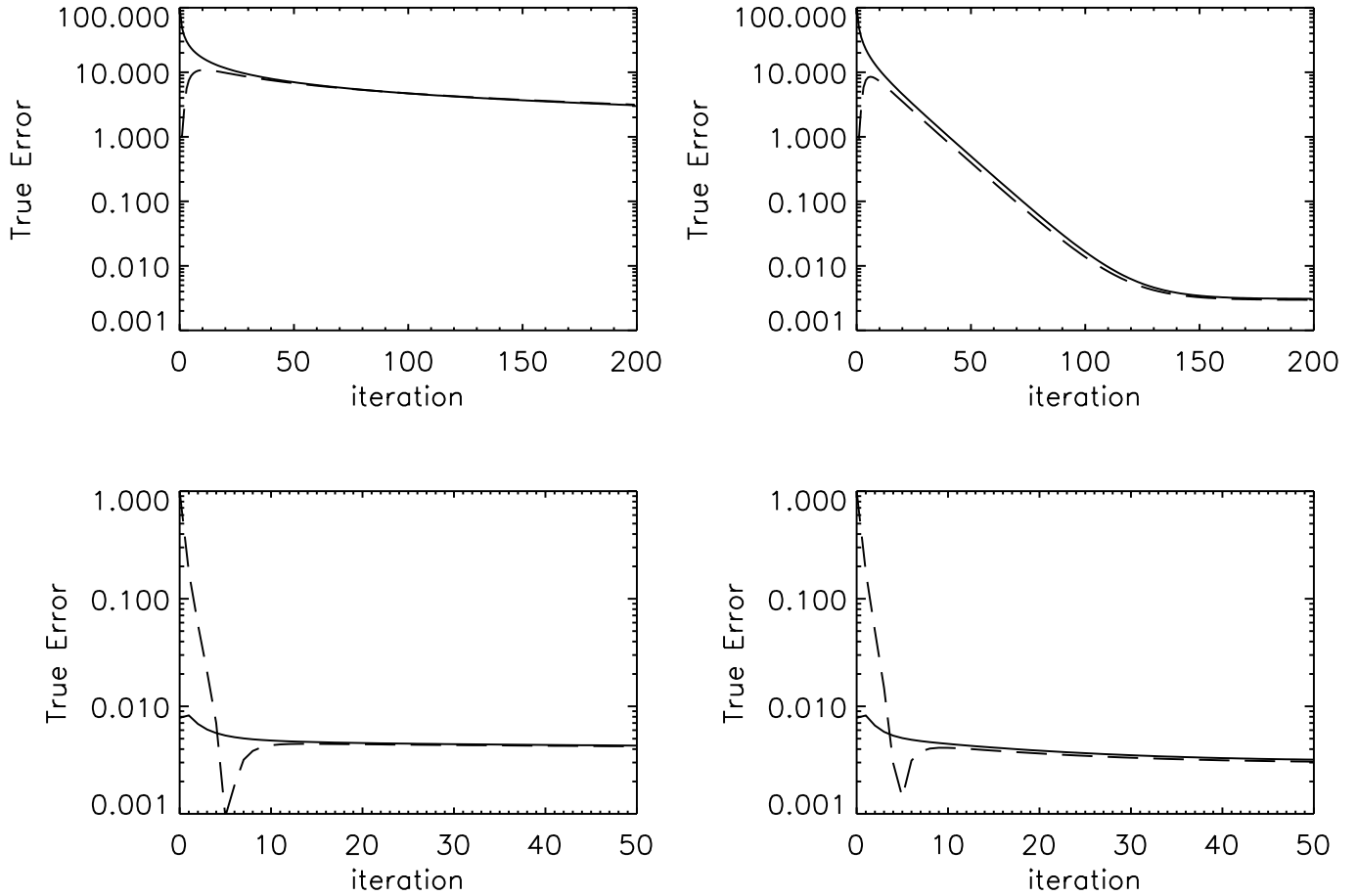


FIG. 2.—Surface true error of ρ_0^0 (solid line) and of ρ_0^2 (dashed line) for a Λ -iteration calculation (left-hand panels) and for a calculation performed with the Jacobi-based ALI method presented in § 3.1 (right-hand panels). The two upper panels correspond to calculations initialized with the LTE density matrix values, while the two lower panels correspond to calculations initialized using the self-consistent ρ -values of the case in which polarization phenomena are neglected.

along each three consecutive grid points that we denote with the symbols M , O , and P , O being the grid point of interest. If we use the symbol X to denote either the Stokes I or Q parameter, the SC method states that X at a given interior grid point O , for a given frequency and ray direction, can be expressed as

$$X_O = X_M e^{-\Delta\tau_M} + \Psi_M S_M + \Psi_O S_O + \Psi_P S_P, \quad (16)$$

where $\Delta\tau_M$ is the optical distance on segment MO , and the Ψ -quantities are functions of the optical distances between O and the upwind point M and between O and the downwind point P , while S_k (with $k = M, O$, or P) are the S_I (if $X = I$) or the S_Q (if $X = Q$) source function values at such grid points k . This formula applies to interior grid points. At the boundary grid points the Stokes X -parameter for rays going out of these boundaries is calculated assuming the source function varies linearly along M and O , i.e., applying a formula like equation (16), but with $\Psi_P = 0$.

This formal solution technique has been very useful for developing two- and three-dimensional multilevel transfer methods (see Fabiani Bendicho, Trujillo Bueno, & Auer 1997; Fabiani Bendicho & Trujillo Bueno 1999; and references therein). The SC method can also be generalized for solving the full Stokes vector transfer equation including magneto-optical effects (see Trujillo Bueno 1998). In fact,

this was the formal solver used by Trujillo Bueno & Landi Degl'Innocenti (1996a) in their investigation of the non-LTE Zeeman line transfer problem. In the present case of resonance line polarization we have used the standard SC method with parabolic accuracy in order to solve the scalar transfer equations (3) and (4). Our SC formal solver calculates, at all the spatial grid points, the I and Q parameters for each frequency and direction. To this end, one has to evaluate first S_I and S_Q from the current values of the density matrix elements. This information allows the calculation of the radiation field tensors $\bar{J}_0^0$ and $\bar{J}_0^2$ at each grid point i (cf. eqs. [10] and [11]). The operations required to perform this numerical calculation of $\bar{J}_0^0$ and $\bar{J}_0^2$ can be indicated as follows:

$$\begin{aligned} \bar{J}_0^0(i) = & \sum_{j=1}^{NP} \Lambda_0^0(i, j) S_0^0(j) + \sum_{j=1}^{NP} \Lambda_2^0(i, j) S_0^2(j) \\ & + \sum_{j=1}^{NP} \Lambda_c^0(i, j) S_c(j) + T_0^0(i), \end{aligned} \quad (17)$$

$$\begin{aligned} \bar{J}_0^2(i) = & \sum_{j=1}^{NP} \Lambda_0^2(i, j) S_0^0(j) + \sum_{j=1}^{NP} \Lambda_2^2(i, j) S_0^2(j) \\ & + \sum_{j=1}^{NP} \Lambda_c^2(i, j) S_c(j) + T_0^2(i), \end{aligned} \quad (18)$$

where

$$\Lambda_0^0(i, j) = \frac{1}{2} \int dx \phi_x(i) r_x(i) \int_{-1}^1 d\mu \Lambda_{x\mu}(i, j), \quad (19)$$

$$\Lambda_2^0(i, j) = \frac{w_{J_u J_l}^{(2)}}{4\sqrt{2}} \int dx \phi_x(i) r_x(i) \int_{-1}^1 d\mu (3\mu^2 - 1) \Lambda_{x\mu}(i, j), \quad (20)$$

$$\Lambda_0^2(i, j) = \Lambda_2^0(i, j) / w_{J_u J_l}^{(2)}, \quad (21)$$

$$\Lambda_2^2(i, j) = \frac{w_{J_u J_l}^{(2)}}{16} \int dx \phi_x(i) r_x(i) \times \int_{-1}^1 d\mu [(3\mu^2 - 1)^2 + 9(\mu^2 - 1)^2] \Lambda_{x\mu}(i, j), \quad (22)$$

$$\Lambda_c^0(i, j) = \frac{1}{2} \int dx \phi_x(i) [1 - r_x(i)] \int_{-1}^1 d\mu \Lambda_{x\mu}(i, j), \quad (23)$$

$$\Lambda_c^2(i, j) = \frac{1}{4\sqrt{2}} \int dx \phi_x(i) [1 - r_x(i)] \times \int_{-1}^1 d\mu (3\mu^2 - 1) \Lambda_{x\mu}(i, j), \quad (24)$$

and where $T_0^0(i)$ and $T_2^0(i)$ are given by expressions similar to equations (10) and (11), respectively, but using $T_{x\Omega}^I$ instead of $I_{x\Omega}$ and $T_{x\Omega}^Q$ instead of $Q_{x\Omega}$.

If we used the current “old” values of S_0^0 and S_0^2 at all the spatial grid points when calculating formally the radiation field tensors according to equations (17) and (18), the result would simply be $\bar{J}_0^0(i) = \bar{J}_0^0(i)^{\text{old}}$ and $\bar{J}_0^2(i) = \bar{J}_0^2(i)^{\text{old}}$. As mentioned above, the Λ -iteration method is obtained when these “old” radiation field tensors are introduced into the rate equations (1) and (2) with the aim of obtaining “new” estimates of the density matrix elements at each spatial grid point independently, as indicated by equations (12) and (13).

In order to improve the convergence properties of this method, but still keeping its simplicity, we introduce into equations (17) and (18) the known “old” values of the ρ -elements at all the grid points except at the grid point i being considered. At this grid point i we just write S_0^{new} and $S_0^{2\text{new}}$ implicitly, since we still do not know their particular numerical values. As a result we obtain

$$\bar{J}_0^0(i) = \bar{J}_0^0(i)^{\text{old}} + \Lambda_0^0(i, i) \delta S_0^0(i) + \Lambda_2^0(i, i) \delta S_0^2(i), \quad (25)$$

$$\bar{J}_0^2(i) = \bar{J}_0^2(i)^{\text{old}} + \Lambda_0^2(i, i) \delta S_0^0(i) + \Lambda_2^2(i, i) \delta S_0^2(i), \quad (26)$$

where $\delta S_0^K(i) = S_0^{K\text{new}} - S_0^{K\text{old}}$.

Introducing these expressions now into the SE equations (1) and (2), we finally obtain a system of two equations with $\delta S_0^0(i)$ and $\delta S_0^2(i)$ as unknowns:

$$[1 - (1 - \epsilon) \Lambda_0^0(i, i)] \delta S_0^0 = (1 - \epsilon) \bar{J}_0^0(i)^{\text{old}} + \epsilon B_v - S_0^0(i)^{\text{old}} + (1 - \epsilon) \Lambda_2^0(i, i) \delta S_0^2, \quad (27)$$

$$\begin{aligned} & \left[1 - \mathcal{H}^{(2)} \frac{1 - \epsilon}{1 + \delta^{(2)}(1 - \epsilon)} \Lambda_2^2(i, i) \right] \delta S_0^2 \\ &= \mathcal{H}^{(2)} \frac{1 - \epsilon}{1 + \delta^{(2)}(1 - \epsilon)} w_{J_u J_l}^{(2)} \bar{J}_0^2(i)^{\text{old}} - S_0^2(i)^{\text{old}} \\ &+ \mathcal{H}^{(2)} \frac{1 - \epsilon}{1 + \delta^{(2)}(1 - \epsilon)} \Lambda_0^2(i, i) \delta S_0^0. \end{aligned} \quad (28)$$

The solution of this system of equations gives the corrections at a given iterative step. We can consider this iterative scheme as a generalization, to the standard two-level atom

resonance line polarization problem, of Jacobi’s operator splitting method. Its application only requires the evaluation of the four diagonal operators given by equations (19)–(22) and to perform, at each iterative step, formal solutions of two independent transfer equations (cf. eqs. [3] and [4]) in order to be able to calculate $\bar{J}_0^0(i)^{\text{old}}$ and $\bar{J}_0^2(i)^{\text{old}}$ from the current “old” values of the density matrix elements. The numerical calculation of the diagonal elements of the four operators Λ_Q^K proceeds as simply as with the evaluation of $\Lambda_0^0(i, i)$ in the standard unpolarized transfer case (see, e.g., Olson & Kunasz 1987), with the only novelty that we have now, in general, four operators defined by equations (19)–(22). An important point to keep in mind is that our SC formal solver routine itself calculates simultaneously both the two radiation field tensors $\bar{J}_Q^K$ from the “old” values of the density matrix elements and the diagonal elements of the four operators Λ_Q^K . This is done with a computational work that scales *linearly* with the number of spatial grid points NP , and *linearly* with the number of frequencies and ray directions.

We point out that equations (12) and (13) of the Λ -iteration method can be formally obtained from equations (27) and (28) by omitting here the terms containing explicitly the diagonal operators Λ_Q^K , i.e., by making $\Lambda_0^0(i, i) = \Lambda_2^2(i, i) = \Lambda_0^2(i, i) = \Lambda_2^0(i, i) = 0$. The Jacobi-type method that we have just presented in equations (27) and (28) is characterized by the appearance of three *independent* Λ -operators. Other similar Jacobi-based methods can be obtained by forcing one or two of such diagonal operators to be equal to zero, while still retaining the other(s). To gain some insight as to which operators one could force to be zero, we show in Figure 3 the variation with height of these diagonal operators. As expected from their defining equations (cf. eqs. [19]–[22]), $|\Lambda_0^0(i, i)| > |\Lambda_2^2(i, i)| \gg |\Lambda_0^2(i, i)| = |\Lambda_2^0(i, i) / w_{J_u J_l}^{(2)}|$.

Thus, a first choice is to make only $\Lambda_2^0(i, i) = \Lambda_0^2(i, i) = 0$, i.e., to neglect the last right-hand side terms of equations (27) and (28). This implies that, at each iterative step, one first solves equation (27) to obtain $S_0^{\text{new}} = S_0^{\text{old}} + \delta S_0^0$, and subsequently one solves equation (28) to find $S_0^{2\text{new}}$.

A second choice consists in making $\Lambda_2^2(i, i) = \Lambda_0^2(i, i) = \Lambda_2^0(i, i) = 0$. This implies that the calculation of the δS_0^0 corrections are obtained by a Jacobi-type iteration given by equation (27), but with its last right-hand side term neglected. The resulting equation that gives the δS_0^2 correction is here formally similar to that given by the Λ -iteration method (see eq. [13]), but with the crucial difference that $\bar{J}_0^2$ is now being improved at the rate given by the previous equation that yields S_0^0 . The reason for this is that $\bar{J}_0^2$ is dominated by the Stokes I parameter, and this intensity is, in turn, basically set by the values of S_0^0 .

Figure 4 shows the variation of the maximum relative change (R_c) versus the iteration number for these three Jacobi-based iterative methods. The conclusion is that the solution of standard resonance line polarization problems using methods based on Jacobi iteration can simply be done making use only of the information provided by the diagonal operator Λ_0^0 . The physical explanation of this result has just been given in the last paragraph, and it constitutes an important point to keep in mind.

3.2. Methods Based on Gauss-Seidel Iteration

We turn now to describing how the Gauss-Seidel-based methods of Trujillo Bueno & Fabiani Bendicho (1995) can

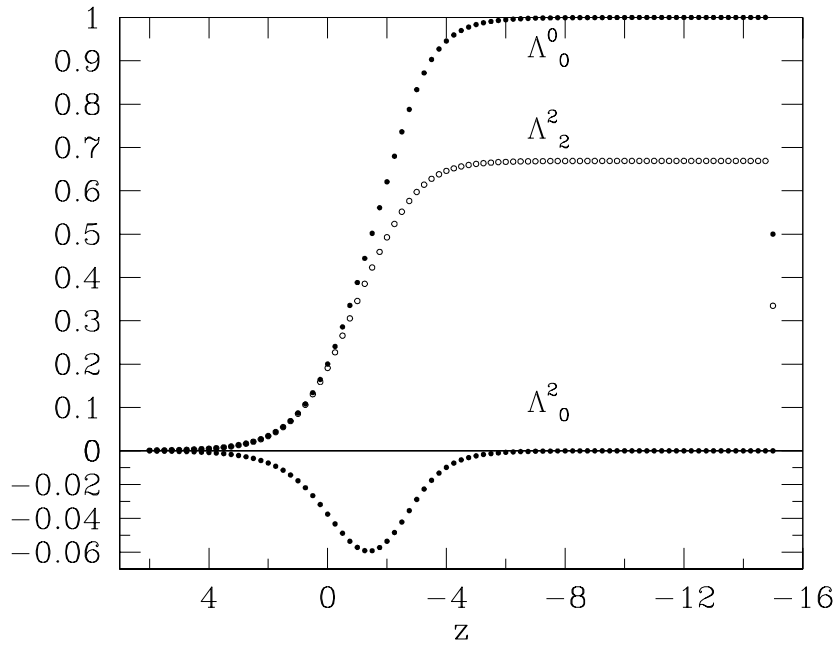


FIG. 3.—Variation with atmospheric height of the diagonal elements of the Λ_0^0 , Λ_0^2 , and Λ_2^2 operators. Note that the vertical scale changes between positive and negative values. The height z is measured in units of the opacity scale height.

be suitably generalized for solving resonance line polarization problems. The idea consists of introducing into equations (17) and (18) the known “old” values of S_0^0 and S_0^2 at the spatial grid points $i + 1, i + 2, \dots, NP$, while at the grid points $1, 2, \dots, i$ we just write $S_0^{0\text{new}}$ and $S_0^{2\text{new}}$. As a result we obtain

$$\bar{J}_0^0(i) = \bar{J}_0^0(i)^{\text{old and new}} + \Lambda_0^0(i, i)\delta S_0^0(i) + \Lambda_2^0(i, i)\delta S_0^2(i), \quad (29)$$

$$\bar{J}_0^2(i) = \bar{J}_0^2(i)^{\text{old and new}} + \Lambda_0^2(i, i)\delta S_0^0(i) + \Lambda_2^2(i, i)\delta S_0^2(i), \quad (30)$$

where $\delta S_Q^K(i) = S_Q^K(i)^{\text{new}} - S_Q^K(i)^{\text{old}}$ (with $Q = 0$ and $K = 0$ and 2), and $\bar{J}_Q^K(i)^{\text{old and new}}$ are the two radiation field tensors calculated via formal solutions of the I and Q transfer equations, but using the “new” values of S_Q^K previously obtained at grid points $1, 2, \dots, i - 1$, and the “old” values at $i, i + 1, \dots, NP$ (see below and also the paper by Trujillo Bueno & Fabiani Bendicho 1995 for details).

Introducing these expressions now into the SE equations, we finally obtain a system of two equations with $\delta S_0^0(i)$ and $\delta S_0^2(i)$ as unknowns. This system of two equations with two unknowns is identical to that of equations (27) and (28), but with $\bar{J}_Q^K(i)^{\text{old and new}}$ instead of $\bar{J}_Q^K(i)^{\text{old}}$. Since, as concluded above, the neglect of the diagonal operators $\Lambda_0^0(i, i)$, $\Lambda_2^0(i, i)$, and $\Lambda_2^2(i, i)$ does not cause the convergence rate to deteriorate, we give here the simplified system of equations that can easily be solved at each iterative step to yield our Gauss-Seidel corrections:

$$[1 - (1 - \epsilon)\Lambda_0^0(i, i)]\delta S_0^0 = (1 - \epsilon)\bar{J}_0^0(i)^{\text{old and new}} + \epsilon B_\nu - S_0^{0\text{old}}, \quad (31)$$

$$\delta S_0^2 = \mathcal{H}^{(2)} \frac{1 - \epsilon}{1 + \delta^{(2)}(1 - \epsilon)} w_{J_u J_l}^{(2)} \bar{J}_0^2(i)^{\text{old and new}} - S_0^{2\text{old}}. \quad (32)$$

It is very important to clarify the meaning of these equa-

tions, whose solution gives the Gauss-Seidel corrections at a given iterative step:

1. First, at point $i = 1$ (which one can freely assign to any of the two boundaries of the medium under consideration) use “old” density matrix element values to calculate $\bar{J}_Q^K(1)$ via a formal solution. Solve the above system of equations (31) and (32) to calculate $S_Q^K(1)^{\text{new}}$.
2. Go to the next point $i = 2$ and use $S_Q^K(1)^{\text{new}}$ and the “old” values $S_Q^K(j)^{\text{old}}$ at points $j = 2, 3, \dots, NP$ to get $\bar{J}_Q^K(2)$ via a formal solution. Solve the system of equations (31) and (32) to calculate $S_Q^K(2)^{\text{new}}$.
3. Go to the next spatial point k and use the previously obtained “new” ρ -values at $j = 1, 2, \dots, k - 1$, but the still “old” ones at $j = k, k + 1, \dots, NP$ to get $\bar{J}_Q^K(k)$ via a formal solution and $S_Q^K(k)^{\text{new}}$ as dictated by the system of equations (31) and (32).
4. Go to the next point $k + 1$ and repeat what has just been indicated in the previous point until arriving to the other boundary point.

The result of what we have just indicated is a pure GS iteration. A SOR iteration is obtained by doing the corrections as follows:

$$\delta S_Q^K(i)^{\text{SOR}} = \omega \delta S_Q^K(i)^{\text{GS}}, \quad (33)$$

where ω is a parameter with an optimal value between 1 and 2 that can be found easily (see Trujillo Bueno & Fabiani Bendicho 1995).

Figure 5 shows an example of the convergence rate of all these iterative methods applied to this resonance line polarization problem. We chose a constant property atmosphere with a collisional destruction probability $\epsilon = 10^{-4}$. We point out that the computing time *per iteration* is similar for all these methods and that matrix inversions are not performed at all. Thus, the important point to keep in mind is

that our pure GS method is 4 times faster than the Jacobi method (i.e., than ALI), while our SOR method for polarization radiative transfer applications is 10 times faster. Our implementation of our GS method is 4 times faster than Jacobi, instead of just twice as fast, as one might expect from a reading of Hageman & Young's (1981) book, because here we have implemented the GS strategy briefly mentioned in the conclusions of the paper by Trujillo Bueno & Fabiani Bendicho (1995), i.e., we have developed a SC formal solver such that each call to it produces as an output two truly GS or SOR iterations: one after the *incoming* pass and another after the *outgoing* pass. A detailed description of this improved Gauss-Seidel-based iterative scheme will be given elsewhere.

Figure 6 is aimed at demonstrating why it is important to have access to iterative schemes that converge much faster than a Jacobi-based one. The figure shows examples of the performance of the Jacobi-based method discussed above as applied to the standard resonance line polarization problem in isothermal atmospheres. The four left-hand panels give the variation of the maximum relative change (R_c) with the iteration number, while the four right-hand panels show the relative *true* error (T_e) at the surface grid point. This is shown for spatial grids of increasing spatial resolution, ranging from 4.5 points per decade to 36 points per decade.

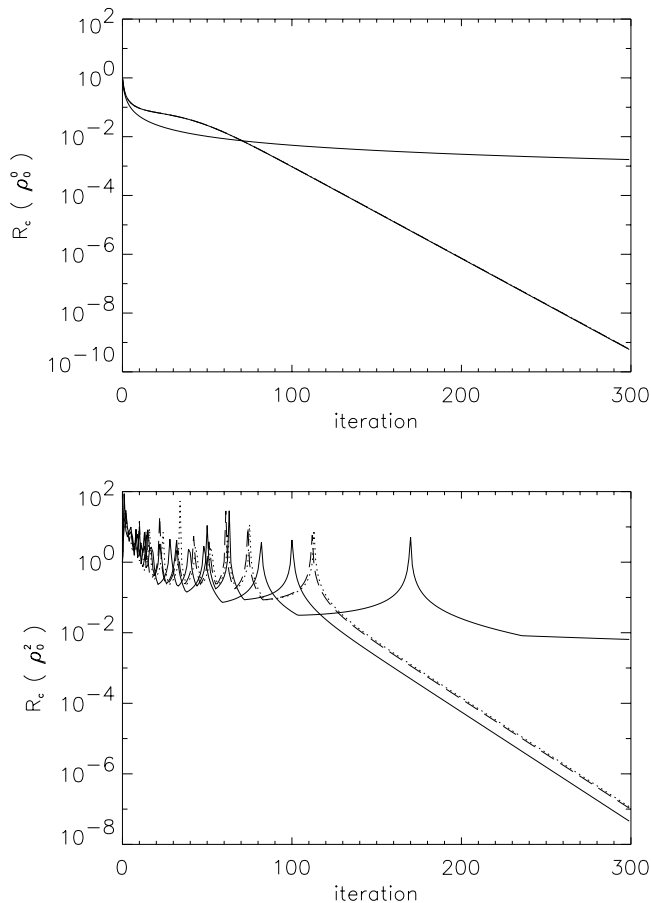


FIG. 4.—Variation with the iteration number of the maximum relative change (R_c) for the ρ_0^0 and ρ_0^2 density matrix elements in a Λ -iteration calculation (solid thin lines), and in Jacobi-based iterative schemes. The solid thick lines correspond to the case in which the four diagonal operators are taken into account, the dashed lines to the case in which $\Lambda_0^0(i, i) = \Lambda_0^2(i, i) = 0$, and the dotted lines to the case in which only the diagonal operator $\Lambda_0^0(i, i)$ is retained. Note that the $R_c(\rho_0^0)$ in the last three cases are indistinguishable.

Within each R_c panel the upper curve corresponds to the alignment coefficient (ρ_0^2) and the lower one to the overall population of the upper level (ρ_0^0). In all these calculations the initialization was done using the LTE ρ -values. Note that the finer the grid, the slower the convergence rate. An additional important feature to note is that in order to obtain the self-consistent ρ -values with a true error smaller than 0.1%, it is necessary to iterate about 350 times using a spatial grid with 18 points per decade. Although this computational work can be reduced by a factor of 2 via the application of the acceleration technique of Ng (1974), it remains a sizable computing effort. However, using our SOR method we only need to iterate 30 times in order to reach the accuracy that the chosen grid can provide. With our pure GS method combined with Ng acceleration, we need to iterate 50 times. This means a dramatic improvement over the Jacobi-based ALI method because the cost per iteration is virtually the same in all these iterative schemes.

4. SOME ILLUSTRATIVE EXAMPLES

In this section we aim at showing some results that demonstrate the accuracy of our methods and give information about the variation with height of the density matrix elements.

Figure 7 shows how the exact solution to the resonance line polarization problem in isothermal atmospheres is reached iteratively using our SOR method. To this end the figure gives the variation with line-integrated optical depth of the quantity $[(S_0^0)^2 + (S_0^2)^2]^{1/2}$. The surface value of this quantity must be equal to $\sqrt{\epsilon} B_\nu$, according to the $\sqrt{\epsilon}$ law (Ivanov 1990; Landi Degl'Innocenti & Bommier 1994; Frisch 1998). This is, in fact, the result that is obtained from the iterative calculation, as shown in the figure. Table 1 gives, with 5 significant digits, the surface value of $[(S_0^0)^2 + (S_0^2)^2]^{1/2}$ for several values of the non-LTE parameter ϵ and for calculations in five spatial grids of increasing resolution. In all these cases a very fine discretization in frequency and direction was used.

Figure 8 gives information about the variation with height of ρ_0^2/ρ_0^0 for calculations in isothermal atmospheres with $\epsilon = 10^{-4}$ and for several values of the depolarizing rate. This figure shows that, even with values of the depolar-

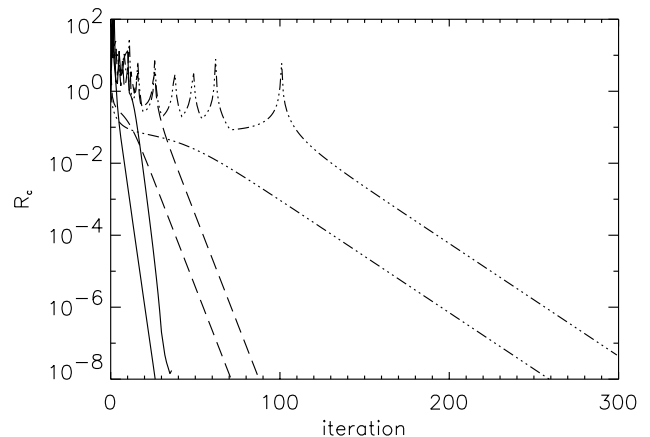


FIG. 5.—Maximum relative change R_c vs. the iteration number for different iterative schemes. Dash-dotted lines: Jacobi-based method. Dashed lines: Pure Gauss-Seidel (GS) method. Solid lines: Successive overrelaxation (SOR) method (with $\omega = 1.6$). For each pair of lines the upper one gives $R_c(\rho_0^2)$, and the lower one gives $R_c(\rho_0^0)$.

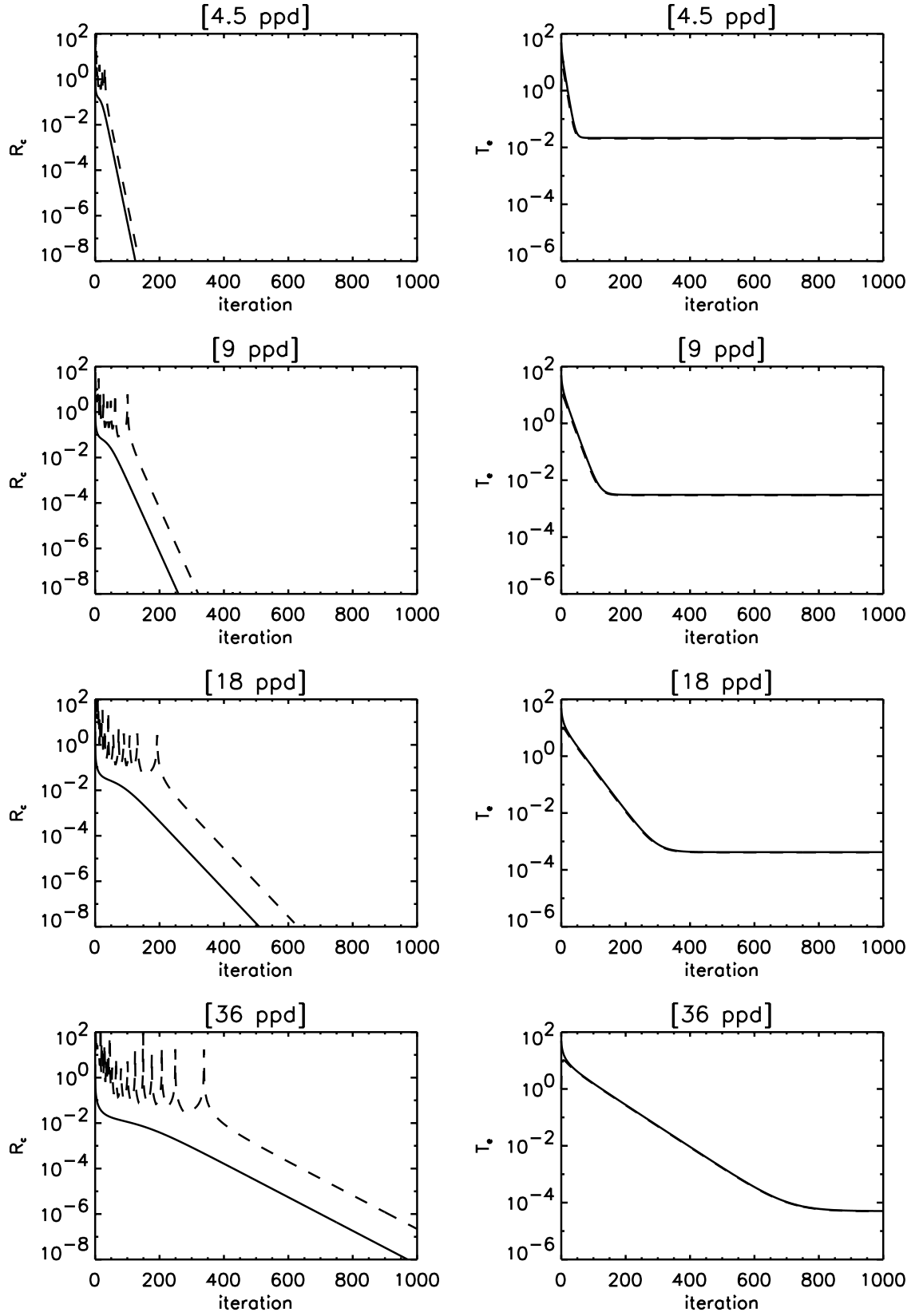


FIG. 6.—Variation with the iteration number of the maximum relative change (R_c) and of the relative true error at the *surface* point (T_e) for the ρ_0^0 (solid lines) and ρ_0^2 (dashed lines) density matrix elements in various spatial grids ranging from 4.5 to 36 points per decade of optical depth.

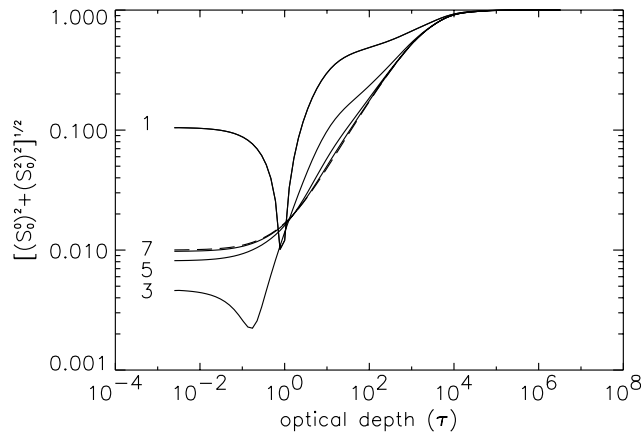


FIG. 7.—Behavior of $[(S_0^0)^2 + (S_0^2)^2]^{1/2}$ in the first iterations of the SOR iterative scheme (with $\omega = 1.6$). The dashed line is the fully converged solution.

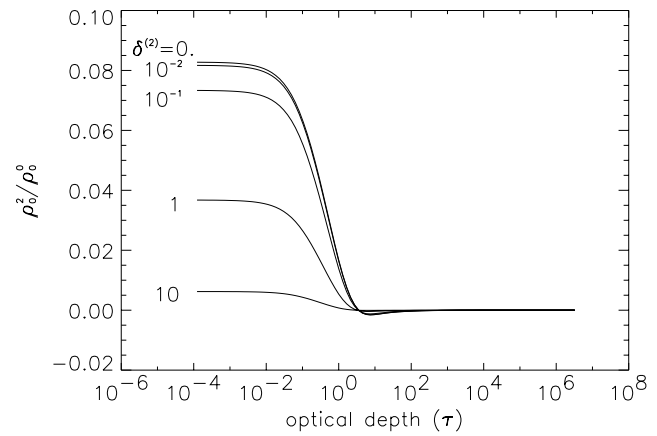


FIG. 8.—Variation with line-integrated optical depth of ρ_0^2/ρ_0^0 in an isothermal atmosphere with $\epsilon = 10^{-4}$ for different depolarizing collisional rates $\delta^{(2)}$.

izing rate of the order of the Einstein A_{ul} coefficient, ρ_0^2/ρ_0^0 is still significant ($\sim 4\%$ at the surface). We point out that for the zero depolarizing rate case the value at the surface is $\rho_0^2/\rho_0^0 = 0.08292$, i.e., 8.3%. This agrees remarkably well with the result reported by Bommier & Landi Degl'Innocenti (1996) (i.e., with $\rho_0^2/\rho_0^0 = 0.08294$), who also solved this resonance line polarization problem, but applying direct methods of solution that require the construction and inversion of large matrices. Tables 2 and 3 provide more detailed information in this respect. Table 2 gives the surface value of ρ_0^2/ρ_0^0 for $\epsilon = 10^{-4}$, for calculations performed in grids of increasing resolution in angles and space, but all of them using a very fine frequency grid. Table 3 shows the relative *true* errors (in percentage) corresponding to the results of Table 2. These calculations were carried out using Gaussian quadrature for the angular integration, and they demonstrate that to perform the angular integration using a numerical quadrature of high resolution is a key point in order to be able to achieve an accuracy better than 0.1%. We always do the frequency integration using the trapezoidal rule. As Figure 9 demonstrates, it suffices to choose two frequency points per Doppler width for calculations in isothermal atmospheres with a Gaussian line profile.

In Figure 10 the emergent fractional linear polarization Q/I is shown for several values of ϵ , for two depolarizing rate cases, and assuming an observation at $\mu = 0.1$. Note that, for the standard two-level atom resonance line polarization problem, the smaller ϵ the larger the emergent fractional linear polarization, independent of whether or not depolarizing collisions come into play. Table 4 gives the emergent fractional linear polarization for decreasing values of ϵ assuming a tangential observation ($\mu = 0$). We point out that, in agreement with Bommier & Landi Degl'Innocenti (1996), we obtain $Q/I = -9.061\%$ for $\epsilon = 10^{-4}$ and $Q/I = -9.441\%$ for $\epsilon \rightarrow 0$. We also find that the limb polarization calculated for the other ϵ -values follows the law $Q/I(\%) = 9.443 - 38.05\epsilon^{1/2}$ given by Ivanov, Grachev, & Loskutov (1997).

Finally, we should mention that we obtained $Q/I = 11.711\%$ for the Rayleigh scattering limit (see Chandrasekhar 1960, who gives 11.713%). This limit can be investigated with our code by choosing one frequency point and by making the line absorption profile equal to unity.

5. CONCLUSIONS

In this paper we have shown how to generalize to polarization transfer the same operator splitting methods that have been so successful for solving unpolarized non-LTE problems. These iterative methods are the Jacobi-based ALI method (see Olson et al. 1986) and the methods based on Gauss-Seidel iteration developed by Trujillo Bueno & Fabiani Bendicho (1995). These GS-based methods are superior to the Jacobi-based ALI method because they yield an order of magnitude of improvement in the total computational work. Our pure GS method is 4 times faster than Jacobi, while our SOR method is 10 times faster. In spite of this very high convergence rate, each iteration basically requires the performing of a formal solution of the transfer equations in order to calculate the radiation field tensors from the current values of the quantities that specify the atomic excitation (i.e., the atomic density matrix elements). In our implementation of all these methods, only local approximate operators come into play, i.e., it is not necessary either to construct or to invert matrices.

The theoretical framework that we have selected for formulating polarization transfer problems is the density matrix theory (see Landi Degl'Innocenti 1983). Although the resonance line polarization case that we have considered in this first paper can also be formulated using some phenomenological theories (see Rees 1978; Stenflo 1994;

TABLE 1
SURFACE VALUE OF $[(S_0^0)^2 + (S_0^2)^2]^{1/2}$ ^a

ϵ	Δz				
	1.0 (2.30)	0.75 (3.07)	0.5 (4.61)	0.25 (9.21)	0.1 (23.03)
10^{-2}	0.10564	0.10251	0.10079	0.10010	0.10000
10^{-4}	0.01154	0.01067	0.01021	0.01003	0.01000
10^{-6}	0.00126	0.00111	0.00103	0.00100	0.00100
10^{-8}	0.00013	0.00011	0.00010	0.00010	0.00010

NOTE.—A very high angular and frequency resolution has been used in all these calculations. Δz is the geometrical grid spacing measured in units of the opacity scale height, with the ensuing number of grid points per decade of line-integrated optical depth indicated in parentheses.

^a Given for four values of the non-LTE parameter ϵ in isothermal atmospheres, and for five spatial grids of increasing resolution.

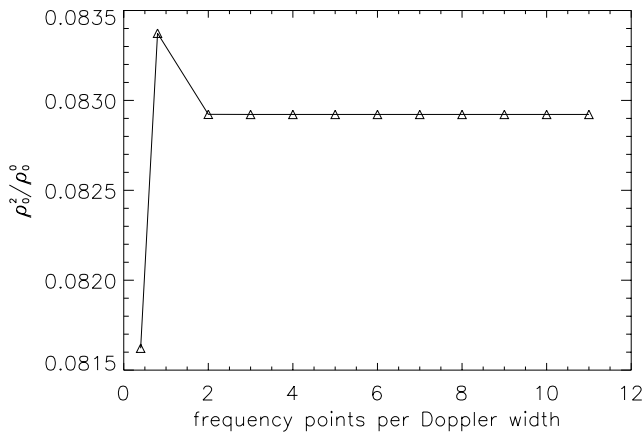


FIG. 9.—Information is given about the accuracy of the trapezoidal rule for the frequency integration. It shows the variation of the surface value of ρ_0^2/ρ_0^0 with the number of frequency points per Doppler width.

TABLE 2
SURFACE VALUE OF ρ_0^2/ρ_0^0 ^a

n_μ	Δz				
	1.0 (2.30)	0.75 (3.07)	0.5 (4.61)	0.25 (9.21)	0.1 (23.03)
3	0.07635	0.07675	0.07696	0.07704	0.07705
5	0.08007	0.08052	0.08076	0.08084	0.08085
7	0.08108	0.08155	0.08178	0.08186	0.08187
11	0.08172	0.08218	0.08242	0.08249	0.08250
15	0.08193	0.08239	0.08262	0.08269	0.08270
21	0.08205	0.08251	0.08273	0.08281	0.08282
37	0.08214	0.08259	0.08281	0.08289	0.08290
41	0.08215	0.08260	0.08282	0.08290	0.08290
51	0.08216	0.08261	0.08283	0.08291	0.08292
57	0.08217	0.08262	0.08284	0.08291	0.08292
61	0.08217	0.08262	0.08284	0.08291	0.08292

NOTE.—An extremely fine frequency grid (with $\Delta x = 0.1\Delta\nu_D$) has always been used integrating up to 5 Doppler widths assuming Gaussian absorption profiles; n_μ is the number of Gaussian points $\mu_i \in [0,1]$ of the angular quadrature, and Δz measures the grid spacing in units of the opacity scale height, with the ensuing number of grid points per decade of line-integrated optical depth indicated in parentheses.

^a Given for several calculations with $\epsilon = 10^{-4}$ and using different angular and spatial resolutions.

Ivanov 1995), the density matrix theory has the advantage of allowing the treatment of more complex polarization problems, like the two-level atom case including the effects of ground-level polarization, or true realistic polarization transfer problems, like the case of scattering polarization with multilevel atomic models in the presence of magnetic fields (see Trujillo Bueno 1999).

In this paper we have also given an indication concerning the solution method that should first be tried for developing a fast computer code for general multilevel polarized radiative transfer. As demonstrated in § 3, the well-known Λ -iteration method is capable of yielding the self-consistent ρ_Q^K density matrix elements if the chosen initialization is given by the self-consistent ρ_0^0 values corresponding to the case in which polarization phenomena are neglected. These ρ_0^0

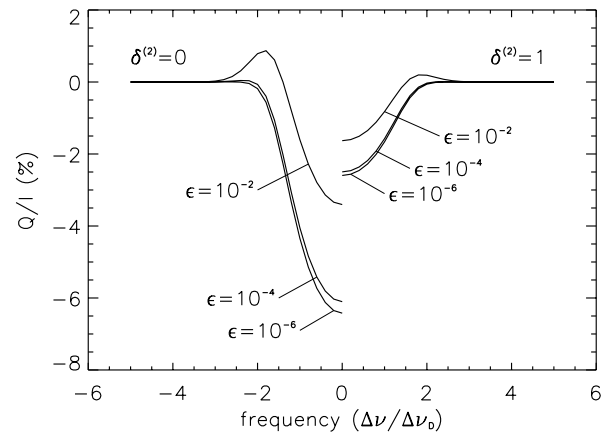


FIG. 10.—Emergent fractional linear polarization Q/I profiles at $\mu = 0.1$ for different ϵ and $\delta^{(2)}$ values. For each depolarizing rate $\delta^{(2)}$ value only half of the Q/I profile is shown. The positive Q -direction is the perpendicular to the tangential to the stellar limb.

TABLE 3
SURFACE TRUE ERRORS (%) OF ρ_0^2/ρ_0^0 VALUES GIVEN IN TABLE 2

n_μ	Δz				
	1.0 (2.30)	0.75 (3.07)	0.5 (4.61)	0.25 (9.21)	0.1 (23.03)
3	7.929	7.449	7.189	7.093	7.081
5	3.444	2.895	2.613	2.515	2.504
7	2.223	1.661	1.378	1.281	1.270
11	1.451	0.890	0.611	0.517	0.507
15	1.202	0.644	0.368	0.276	0.266
21	1.054	0.501	0.229	0.138	0.128
37	0.944	0.398	0.130	0.040	0.030
41	0.934	0.388	0.121	0.031	0.021
51	0.917	0.374	0.107	0.018	0.008
57	0.911	0.369	0.102	0.013	0.003
61	0.908	0.366	0.099	0.010	0.000

NOTE.—The reference value is the one corresponding to a grid with $n_\mu = 61$ and $\Delta z = 0.05$.

values are nothing but the atomic level populations that can be rapidly computed using any of the fast multilevel methods for unpolarized radiative transfer that are now available (see, e.g., Fabiani Bendicho et al. 1997).

In forthcoming papers we will show how the same operator splitting methods considered here can be suitably applied to other linear and nonlinear polarization problems of more complex nature that can be beautifully formulated within the framework of the density matrix theory.

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TABLE 4
SURFACE VALUE OF ρ_0^2/ρ_0^0 AND PERCENTAGE VALUE OF THE EMERGENT FRACTIONAL
LINEAR POLARIZATION Q/I^a

ϵ	10^{-2}	10^{-4}	10^{-6}	10^{-8}	10^{-10}
$(\rho_0^2/\rho_0^0)_{\text{surf}} \dots\dots$	0.05666	0.08292	0.08596	0.08626	0.08629
$(Q/I)_{\mu=0} \dots\dots$	-6.132	-9.061	-9.403	-9.437	-9.441
$(\rho_0^2/\rho_0^0)_{\text{surf}} \dots\dots$	0.06846	0.10184	0.10582	0.10623	0.10627
$(Q/I)_{\mu=0} \dots\dots$	-7.442	-11.206	-11.660	-11.707	-11.711

^a Given for a tangential observation in the scattering line polarization case (first two rows) and in the Rayleigh regime (last two rows).

APPENDIX A

DERIVATION OF STATISTICAL EQUILIBRIUM EQUATIONS INCLUDING THE EFFECT OF A WEAK MICROTURBULENT AND ISOTROPIC MAGNETIC FIELD

We consider a two-level atom under the assumptions stated in § 2. The contribution of the radiative rates to the temporal variation of the density matrix element ρ_Q^K (normalized to the overall population of the lower level) in a reference system whose z -axis is directed along the magnetic field is (see Landi Degl'Innocenti 1985)

$$\left[\frac{d}{dt} \rho_Q^K \right]_{\text{rad}} = -iA_{ul} \gamma Q \rho_Q^K - A_{ul} \rho_Q^K + B_{lu} (2J_u + 1)^{-1/2} w_{J_u J_l}^{(K)} (-1)^Q \bar{J}_{-Q}^K, \quad (\text{A1})$$

where $\gamma = 2\pi\nu_L g_{J_u}/A_{ul}$, with ν_L the Larmor frequency, g_{J_u} the Landé factor, A_{ul} and B_{lu} the Einstein coefficients, and $w_{J_u J_l}^{(K)}$ a numerical factor depending on the transition (see Table 1 in Landi Degl'Innocenti 1984) and whose value is unity when $J_l = 0$ and $J_u = 1$. The radiation field tensor J_Q^K is given by a weighted frequency and angular average of the Stokes parameters ($I_{x\Omega}$, $Q_{x\Omega}$, $U_{x\Omega}$, $V_{x\Omega}$):

$$\bar{J}_Q^K = \int dx \phi_x \oint \frac{d\Omega}{4\pi} [\mathcal{T}_Q^K(I_{x\Omega}, \Omega) I_{x\Omega} + \mathcal{T}_Q^K(Q_{x\Omega}, \Omega) Q_{x\Omega} + \mathcal{T}_Q^K(U_{x\Omega}, \Omega) U_{x\Omega} + \mathcal{T}_Q^K(V_{x\Omega}, \Omega) V_{x\Omega}], \quad (\text{A2})$$

where $\mathcal{T}_Q^K$ are the polarization tensors defined in Landi Degl'Innocenti (1984).

On the other hand, since the lower level is assumed to be unpolarized, and as elastic and inelastic collisions relax all the irreducible tensor components independently when the perturbers' velocity is isotropic, the collisional contribution to the rate of change of the ρ_Q^K elements is given by (see Landi Degl'Innocenti et al. 1990)

$$\left[\frac{d}{dt} \rho_Q^K \right]_{\text{coll}} = (2J_u + 1)^{-1/2} C_{lu} \delta_{K0} \delta_{Q0} - C_{ul} \rho_Q^K - D^{(K)} \rho_Q^K, \quad (\text{A3})$$

where C_{lu} and C_{ul} are the standard inelastic collisional rates, and $D^{(K)}$ is the elastic depolarizing collisional rate, which is obviously zero for $K = 0$. (The $(2J_u + 1)^{-1/2}$ factor has been introduced to make the definitions compatible with the standard unpolarized non-LTE problem.) Under the impact approximation hypothesis, the collisional rate equations can be simply added to the radiative ones (i.e., $d\rho_Q^K/dt = [d\rho_Q^K/dt]_{\text{rad}} + [d\rho_Q^K/dt]_{\text{coll}}$). Assuming a Maxwellian velocity distribution for the perturbers, and using the notation $\epsilon = C_{ul}/(A_{ul} + C_{ul})$, $\delta^{(K)} = D^{(K)}/A_{ul}$, and the Einstein relations, the full SE equations (i.e., $d\rho_Q^K/dt = 0$) can be written as

$$(1 + i\Gamma Q)[1 + \delta^{(K)}(1 - \epsilon)] \frac{2h\nu^3}{c^2} \frac{2J_l + 1}{\sqrt{2J_u + 1}} \rho_Q^K = (1 - \epsilon) w_{J_u J_l}^{(K)} (-1)^Q \bar{J}_{-Q}^K + \epsilon B_v \delta_{K0} \delta_{Q0}, \quad (\text{A4})$$

where

$$\Gamma = \gamma \frac{1 - \epsilon}{1 + \delta^{(K)}(1 - \epsilon)}. \quad (\text{A5})$$

In order to express these SE equations in a *new* reference system with the z -axis lying along the vertical, and the magnetic field having polar and azimuthal angles θ_B and φ_B , we consider a rotation R_B bringing the previous (*old*) reference system into this *new* one:

$$R_B = (\alpha, -\theta_B, -\varphi_B), \quad (\text{A6})$$

α being an angle that can be arbitrarily set equal to zero (it is related to the orientation of the x - and y -axes of the two systems). In a change of reference system the statistical tensors transform according to (Landi Degl'Innocenti et al. 1990):

$$[\rho_Q^K]_{\text{new}} = \sum_{Q'} [\rho_{Q'}^K]_{\text{old}} \mathcal{D}_{Q'Q}^{K*}(R_B), \quad (\text{A7})$$

$$[\bar{J}_Q^K]_{\text{new}} = \sum_{Q'} [\bar{J}_{Q'}^K]_{\text{old}} \mathcal{D}_{Q'Q}^{K*}(R_B), \quad (\text{A8})$$

with $\mathcal{D}_{Q'Q}^{K*}(R_B)$ the ordinary rotation matrices (see, e.g., Brink & Satchler 1968). Multiplying equation (A4) by $\mathcal{D}_{Q'Q}^{K*}(R_B)$ and summing over Q , after some simple algebraic manipulations the SE equations in the *new* reference system read

$$[1 + \delta^{(K)}(1 - \epsilon)] \frac{2h\nu^3}{c^2} \frac{2J_l + 1}{\sqrt{2J_u + 1}} [\rho_Q^K]_{\text{new}} = \sum_{Q'} \frac{1}{1 + i\Gamma Q'} (1 - \epsilon) w_{J_u J_l}^{(K)} (-1)^{Q'} [\bar{J}_{-Q'}^K]_{\text{old}} \mathcal{D}_{Q'Q}^{K*}(R_B) + \epsilon B_v \delta_{K0} \delta_{Q0}, \quad (\text{A9})$$

where

$$[\bar{J}_{-Q'}^K]_{\text{old}} = \sum_{Q''} [\bar{J}_{-Q''}^K]_{\text{new}} \mathcal{D}_{-Q''-Q'}^K(-R_B) = \sum_{Q''} [\bar{J}_{-Q''}^K]_{\text{new}} (-1)^{-Q''+Q'} \mathcal{D}_{Q'Q''}^K(R_B). \quad (\text{A10})$$

Substituting equation (A10) into equation (A9) and obviating the “*new*” subindexes, the SE equations in the new reference system become

$$[1 + \delta^{(K)}(1 - \epsilon)] \frac{2h\nu^3}{c^2} \frac{2J_l + 1}{\sqrt{2J_u + 1}} \rho_Q^K = (1 - \epsilon) \sum_{Q'} \frac{1}{1 + i\Gamma Q'} w_{J_u J_l}^{(K)} \sum_{Q''} \bar{J}_{-Q''}^K (-1)^{Q''} \mathcal{D}_{Q'Q''}^K(R_B) \mathcal{D}_{Q'Q}^{K*}(R_B) + \epsilon B_v \delta_{K0} \delta_{Q0}. \quad (\text{A11})$$

In the isotropic microturbulent magnetic field limit the magnetic field changes its polarity several times in the photon mean free path. Thus, at each spatial point, we can average the SE equations for all the magnetic field directions. Taking into account Weyl's theorem,

$$\frac{1}{2\pi} \int_0^{2\pi} \frac{d\alpha}{4\pi} \int_0^\pi \sin \theta_B d\theta_B \int_0^{2\pi} d\varphi_B \mathcal{D}_{Q'Q''}^K(R_B) \mathcal{D}_{Q'Q}^{K*}(R_B) = \frac{1}{2K + 1} \delta_{QQ'}, \quad (\text{A12})$$

the magnetic field angle-averaged SE equations read

$$[1 + \delta^{(K)}(1 - \epsilon)] \frac{2h\nu^3}{c^2} \frac{2J_l + 1}{\sqrt{2J_u + 1}} \rho_Q^K = (1 - \epsilon) \mathcal{H}^{(K)} w_{J_u J_l}^{(K)} (-1)^Q \bar{J}_{-Q}^K + \epsilon B_v \delta_{K0} \delta_{Q0}, \quad (\text{A13})$$

where the Hanle depolarization factor $\mathcal{H}^{(K)} = (2K + 1)^{-1} \Sigma_Q (1 + i\Gamma Q')^{-1}$ has the following explicit values:

$$\mathcal{H}^{(0)} = 1, \quad (\text{A14})$$

$$\mathcal{H}^{(1)} = \frac{1}{3} \left(1 + \frac{2}{1 + \Gamma^2} \right), \quad (\text{A15})$$

$$\mathcal{H}^{(2)} = \frac{1}{5} \left(1 + \frac{2}{1 + \Gamma^2} + \frac{2}{1 + 4\Gamma^2} \right). \quad (\text{A16})$$

At this stage it is of interest to mention that Stenflo (1982) and Landi Degl'Innocenti (1985) found that the scattering phase matrix in a microturbulent and isotropic magnetic field can be obtained from the scattering phase matrix of the nonmagnetic case by simply multiplying some quantities by a factor similar to $\mathcal{H}^{(K)}$.

Particularizing to $K = 0$ and $K = 2$, one obtains equations (1) and (2) of § 2. We point out that the radiation field tensor $\bar{J}_0^1$ is given by a frequency-angle average of the Stokes V -parameter only, that the V -component of the line source function (i.e., S_V^{line}) depends only on ρ_0^1 , and that S_I^{line} and S_Q^{line} are given by equations (8) and (9), respectively (see Landi Degl'Innocenti 1984). Note also, from the SE equation found for $K = 1$, that ρ_0^1 is proportional to $\bar{J}_0^1$, and that atomic orientation (i.e., nonzero values of ρ_0^1) cannot be generated either from the population ρ_0^0 or from the atomic alignment ρ_0^2 . (We point out that, for instance, for a level with $J = 1$ one has $\rho_0^1 = (n_1 - n_{-1})/\sqrt{2}$). Since we are assuming that the radiation illuminating the boundaries has no circular polarization, one reaches the conclusion that $\bar{J}_0^1 = 0$. Therefore, the atomic orientation is given by $\rho_0^1 = 0$ as mentioned in § 2.

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